

AstroMagic v1.5

Photoshop Plugin for Astrophotography

Complete User Manual

March 2026

<https://pimagicstudio.com>

"The AstroMagic tool palette completely automates the repetitive tasks"

Focus your time on the creative steps

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1. Overview

AstroMagic is a Photoshop UXP plugin that completely automates the repetitive tasks of loading TIFF files, organizing them into groups, and creating standard adjustment layers. The result is you're immediately ready to begin the creative processing to make the image your own.

AstroMagic works seamlessly with PiMagic for PixInsight. After PiMagic processes your stacked masters and creates publication-ready TIFFs, AstroMagic imports them into Photoshop with perfect organization — letting you focus entirely on creative enhancement.

1.1 Key Capabilities

Feature	Description
Automatic Import	Load entire folders of TIFF files with intelligent detection of layer types
Smart Grouping	Automatically organizes layers into logical groups: Final, Starless, Stars, Annotations, etc.
Blend Mode Assignment	Sets correct blend modes for each layer type (Screen for stars, Luminosity for L channel)
Palette Support	Full support for SHO (Hubble), HOO, and Ha Enhanced narrowband palettes
Color-Coded Groups	Visual organization with Photoshop label colors matching palette assignments
Adjustment Layers	Auto-adds Levels, Curves, and other adjustment layers with smart defaults
Auto-Endpoints	Automatically calculates and applies optimal black/white points per group
Continuum Subtraction	Built-in Cyan and Red continuum subtraction tools for cleaner narrowband
Structure Enhancement	One-click nebula and star enhancement using Camera Raw filters
Mosaic Support	Handles multi-panel mosaics with per-pane organization
Processing Log	Detailed diagnostic output for troubleshooting and workflow documentation

1.2 What Users Say

"In Pixinsight, it does everything you need to prepare your masters, including changing them all into the TIFF files. And then when you go to Photoshop, you don't have to spend 20 minutes grouping everything — AstroMagic does it for you. It allows you to be creative right out of the gate." — Hawk Wolinski (astrohawk)

"For the first time, I felt like I'd bent time itself: in mere minutes, I produced stunning results that would have otherwise demanded years of trial and error." — Julian Segal

2. Installation & Requirements

2.1 System Requirements

- Adobe Photoshop 2023 (v24.0) or later with UXP plugin support
- macOS 10.15+ or Windows 10/11
- 8GB RAM minimum (16GB+ recommended for large astrophotos)
- Internet connection for license validation

2.2 Installation Steps

Video Tutorial: See “Installing PiMagic” at pimagicstudio.com/about/

1. Download the AstroMagic plugin package from your purchase email
2. Locate your Photoshop plugins folder:
 - **Mac:** /Library/Application Support/Adobe/UXP/Plugins/
 - **Windows:** C:\Program Files\Adobe\Adobe Photoshop [version]\Plug-ins\UXP\
3. Copy the entire AstroMagic folder to the plugins directory
4. Restart Photoshop
5. Open AstroMagic: Tools → AstroMagic
6. Enter your license key when prompted

Note: An Installation Guide PDF is included with the software download, providing detailed step-by-step instructions for Windows, Mac, and Linux.

2.3 Recommended Companion Software

- **PiMagic for PixInsight** — Automates post-integration to create TIFFs for AstroMagic
- **Camera Raw Filter** — Built into Photoshop, used by AstroMagic's enhancement tools

3. Quick Start Guide

Get started in minutes with this basic workflow:

1. **Process with PiMagic:** Run your stacked masters through PiMagic in PixInsight
2. **Open AstroMagic:** Tools → AstroMagic
3. **Select Palette:** Choose SHO, HOO, or Ha Enhanced from dropdown
4. **Import Images:** Click 'Import Images' button
5. **Select Folder:** Navigate to your PiMagic output folder
6. **Wait for Processing:** AstroMagic imports, groups, and configures all layers
7. **Begin Creative Work:** Your document is ready for creative processing!

Tip: AstroMagic detects file types from their names (Final_, Starless_, Stars_, Annotation_) and automatically assigns the correct group, blend mode, and color label.

3.1 What Happens During Import

When you click 'Import Images', AstroMagic performs these automated steps:

- Scans folder for all TIFF files
- Detects layer types from filenames (Final, Starless, Stars, etc.)
- Creates a new Photoshop document
- Imports each TIFF as a layer
- Creates layer groups in the correct stacking order
- Assigns blend modes (Normal, Screen, Luminosity, Color)
- Applies color labels for visual organization
- Adds adjustment layers with auto-calculated endpoints
- Generates a processing log for your records

4. The AstroMagic Panel

The AstroMagic panel is organized into tabs for different aspects of the workflow.

Video Tutorial: See "AstroMagic Panel Overview" at pimagicstudio.com/about/

4.1 Import Tab

The main tab for loading and organizing your images.

Palette Selection

Palette	Description
SHO (Hubble)	S=Red, H=Green, O=Blue — Classic Hubble Space Telescope palette
HOO	H=Red, O=Cyan — Popular two-filter narrowband palette (SII hidden)
Ha Enhanced	H=Red Only — Emphasizes hydrogen-alpha, hides oxygen group

Import Buttons

- **Import Images (Folder):** Select a folder to import all TIFFs
- **Import Images (Files):** Select individual files to import
- **Monitor Folder:** Watch a folder for new files (hot folder mode)

Options

- **Merge Starless as RGB:** Combine separate R, G, B starless layers
- **Enable Debug Logging:** Create detailed processing log

4.2 Continuum Tab

Tools for continuum subtraction to clean narrowband data.

Video Tutorial: See "AstroMagic LRGB+Ha Continuum Subtraction" at pimagicstudio.com/about/

- **Cyan Continuum (Mono):** Remove cyan continuum from monochrome Ha layer
- **Red Continuum (Mono):** Remove red continuum from monochrome OIII layer
- **Cyan Continuum (RGB):** Remove cyan from RGB color image
- **Red Continuum (RGB):** Remove red from RGB color image

4.3 Enhance Tab

Structure enhancement tools using Camera Raw filters.

- **Enhance Nebula:** Apply Camera Raw filter optimized for nebula detail
- **Enhance Stars:** Apply star color enhancement preset

4.4 Groups Tab

Manually create layer groups when needed.

- **Make Sulfur Group:** Create SII group with red label
- **Make Hydrogen Group:** Create Ha group with appropriate palette color
- **Make Oxygen Group:** Create OIII group with blue label
- **Make Annotations Group:** Create annotation overlay group
- **Make RGB Stars Group:** Create stars group for broadband stars
- **Make Global Adjustments:** Create pass-through adjustment group

4.5 Tools Tab

Utility functions and license management.

- **Close All Groups:** Collapse all layer groups for cleaner view
- **Undo:** Undo last operation
- **Load Additional Layers:** Add more TIFFs to existing document
- **Activate/Deactivate License:** Manage license activation

5. Standard Layer Structure

AstroMagic creates a standardized layer structure that puts you immediately in creative mode. Understanding this structure helps you work efficiently.

Video Tutorial: See "AstroMagic Standard Layer Structure" at pimagicstudio.com/about/

5.1 Layer Group Hierarchy

Groups are created in this order from top to bottom:

Group	Blend Mode	Purpose
Final	Normal	Combined final image with stars
Final RGB	Normal	Final RGB combined (broadband)
Final HOO/SHO	Screen	Final narrowband palette output
Final Luminosity	Luminosity	Luminance channel for LRGB
Starless RGB	Normal	Base starless color image
Starless Luminosity	Luminosity	Starless L channel
Starless HOO/SHO	Screen	Starless narrowband combination
Starless Sulfur	Screen	SII starless (red label)
Starless Oxygen	Screen	OIII starless (blue label)
Starless Hydrogen	Screen	Ha starless (green/red by palette)
Global Adjustments	Pass Through	Master adjustment layers
Stars	Screen	Star layers for final compositing
Annotation	Screen	Object labels and overlays

5.2 Blend Mode Logic

AstroMagic assigns blend modes based on layer function:

- **Normal:** Base layers (Final, Starless RGB) that form the foundation
- **Screen:** Additive layers (Stars, narrowband channels, annotations)
- **Luminosity:** Luminance channels that add detail without affecting color
- **Color:** Star layers when combining with narrowband (adds star color)
- **Pass Through:** Global Adjustments group (lets adjustments affect all below)

5.3 Color Labels

Visual organization using Photoshop's label colors:

- **Red:** Sulfur (SII) groups
- **Green:** Hydrogen (Ha) in SHO palette
- **Blue:** Oxygen (OIII) groups, HOO final
- **Violet:** RGB combined and color images
- **Orange:** Final output groups
- **Gray:** Global Adjustments, Annotations

5.4 Adjustment Layers

AstroMagic automatically adds adjustment layers to each group:

- **Levels:** With auto-calculated black and white points
- **Curves:** For contrast and tonal adjustments
- **Hue/Saturation:** For color fine-tuning
- **Color Balance:** For overall color correction

Auto-Endpoints: AstroMagic analyzes each layer's histogram and calculates optimal black and white points, then applies them to the Levels adjustment. This gives you a proper starting point for each channel.

6. Narrowband Palettes

AstroMagic fully supports the three most popular narrowband palettes. Select your palette before importing to ensure correct color assignments.

6.1 SHO (Hubble Palette)

The classic palette used by the Hubble Space Telescope.

- **Sulfur (SII) → Red**
- **Hydrogen (Ha) → Green**
- **Oxygen (OIII) → Blue**

Best for: Targets with strong SII emission (pillars, star-forming regions)

Video Tutorial: See "Hubble Palette SHO" and "AstroMagic SHO" at pimagicstudio.com/about/

6.2 HOO Palette

Popular two-filter palette producing cyan and red tones.

- **Hydrogen (Ha) → Red**
- **Oxygen (OIII) → Green and Blue (Cyan)**
- **Sulfur → Hidden**

Best for: Targets with strong Ha/OIII contrast (Veil, Heart & Soul nebulae)

6.3 Ha Enhanced

Emphasizes hydrogen-alpha signal, ideal for combining with broadband.

- **Hydrogen (Ha) → Red only**
- **Oxygen → Hidden**

Best for: Adding Ha to LRGB data, enhancing emission in broadband images

6.4 Changing Palettes

You can change palettes after import by:

- Selecting a different palette from the dropdown
- Manually adjusting group visibility
- Changing blend modes on narrowband groups

7. Continuum Subtraction

Continuum subtraction removes broadband stellar contamination from narrowband data, producing cleaner emission signals.

Video Tutorial: See "AstroMagic LRGB+Ha Continuum Subtraction" at pimagicstudio.com/about/

7.1 When to Use

- **Red Continuum:** Remove red stellar light from Ha data
- **Cyan Continuum:** Remove cyan/blue stellar light from OIII data

7.2 How It Works

The continuum subtraction tools run Photoshop Actions that:

- Create a duplicate of the broadband channel
- Subtract it from the narrowband using blend modes
- Leave the result ready for further processing

7.3 Mono vs RGB Versions

- **Mono versions:** For processing individual grayscale narrowband masters
- **RGB versions:** For processing combined color images

8. Workflow Examples

8.1 Simple One-Shot Color (OSC)

Input: Final and Starless TIFFs from one-shot color camera

Video: "AstroMagic One-Shot Color" at pimagicstudio.com/about/

1. Process OSC data through PiMagic
2. Open AstroMagic → Click Import Images
3. Select PiMagic output folder
4. AstroMagic creates Final and Starless groups
5. Begin creative adjustments

Result: Clean layer structure with proper blend modes, ready for Curves and color work.

8.2 Dual-Narrowband (OSC + L-Extreme)

Input: OSC clear filter + dual-narrowband filter data

Video: "AstroMagic Dual-Narrowband" at pimagicstudio.com/about/

1. Process both datasets through PiMagic
2. Select HOO palette in AstroMagic
3. Import the combined output folder
4. AstroMagic detects narrowband and OSC layers
5. Blend OSC stars with narrowband starless

Result: Natural star colors from OSC combined with enhanced nebula from narrowband.

8.3 LRGB Processing

Input: L, R, G, B filter data from mono camera

Video: "AstroMagic LRGB" at pimagicstudio.com/about/

1. Process LRGB through PiMagic (creates RGB combination + L)
2. Import into AstroMagic
3. AstroMagic creates Starless RGB and Starless Luminosity groups
4. Luminosity group set to Luminosity blend mode
5. Adjust L intensity with group opacity

Result: Sharp luminance detail combined with smooth RGB color.

8.4 SHO Hubble Palette

Input: Ha, OIII, SII narrowband masters

Video: "AstroMagic SHO" and "Hubble Palette SHO" at pimagicstudio.com/about/

1. Process narrowband through PiMagic with SHO combination
2. Select SHO palette in AstroMagic
3. Import folder
4. Groups created: Starless Sulfur (red), Hydrogen (green), Oxygen (blue)
5. Adjust individual channel curves for color balance

Result: Classic Hubble palette with gold/orange and blue tones.

8.5 LRGB+Ha with Continuum Subtraction

Input: L, R, G, B + Ha filter data

Video: "AstroMagic LRGB+Ha Continuum Subtraction" at pimagicstudio.com/about/

1. Process LRGB+Ha through PiMagic
2. Import into AstroMagic
3. Select the Ha layer
4. Go to Continuum tab → Click Red Continuum (Mono)
5. Blend cleaned Ha with RGB using Screen mode

Result: Clean Ha emission blended with broadband, no red star halos.

8.6 Two-Panel Mosaic (Soul Nebula)

Input: Multi-pane mosaic processed through PiMagic

Video: "AstroMagic Mosaics" at pimagicstudio.com/about/

1. Process mosaic through PiMagic (creates per-pane outputs)
2. Import Pane 1 into AstroMagic
3. Process creatively, save settings
4. Import Pane 2 into new document
5. Apply similar adjustments for consistency
6. Assemble final mosaic in PixInsight or Photoshop

Result: Consistently processed panes ready for seamless assembly.

9. Video Tutorials & Resources

Comprehensive video tutorials are available at pimagicstudio.com/about/ and on the Utah Desert Remote Observatories YouTube channel.

9.1 Getting Started

- **Installing PiMagic** — Windows, Mac and Linux installation guide
- **PiMagic Introduction** — Overview by Craig Stocks
- **Workflow Overview** — Complete PiMagic to AstroMagic pipeline

9.2 AstroMagic Fundamentals

- **AstroMagic Panel Overview** — Tour of all tools and tabs
- **AstroMagic Standard Layer Structure** — Understanding the automatic organization

9.3 Workflow Examples

- **AstroMagic One-Shot Color** — Simple OSC camera workflow
- **AstroMagic Dual-Narrowband** — OSC + L-Extreme combination
- **AstroMagic LRGB** — Mono camera broadband processing
- **AstroMagic SHO** — Hubble palette narrowband
- **AstroMagic LRGB+Ha Continuum Subtraction** — Advanced mono workflow
- **AstroMagic Mosaics** — Multi-panel processing (Soul Nebula example)

9.4 Photoshop Skills

- **The Photoshop Advantage** — Why Photoshop excels for astrophotography
- **Photoshop 101 — Workspace** — Setting up your workspace
- **Photoshop 101 — Layers and Groups** — Essential layer management
- **Photoshop 101 — Advanced Tools** — Blend modes and smart objects

9.5 YouTube Channel

Subscribe to the **Utah Desert Remote Observatories YouTube channel** for tutorials, live streams, and workflow demonstrations:

youtube.com/@utahdesertremoteobserver8328

10. Processing Log

When Debug Logging is enabled, AstroMagic generates a detailed processing log.

10.1 Enabling Debug Logging

- Check "Enable Debug Logging" in the Import tab before importing
- AstroMagic will prompt you to select a save location for the log

10.2 Log Contents

The processing log includes:

- Session start time and settings
- Input files detected with filter classifications
- Document properties (dimensions, color mode, bit depth)
- Groups created with blend modes and label colors
- Layers placed in each group
- Adjustment layers added (Levels, Curves, etc.)
- Auto-endpoint calculations (black/white points)
- Any warnings or errors encountered
- Processing time and summary statistics

Tip: Include the processing log when reporting issues to support — it provides essential diagnostic information.

11. Troubleshooting

Issue	Solution
Plugin not appearing	Verify AstroMagic folder is in correct Plugins/UXP directory. Restart Photoshop.
License error	Check internet connection. Deactivate and reactivate license. Contact support.
Files not detected	Ensure files are TIFF format with standard naming (Final_, Starless_, Stars_)
Wrong groups created	Check file naming matches expected patterns. Use Filter Check in PiMagic.
Blend modes incorrect	Select correct palette before import. Manually adjust if needed.
Import hangs	Large files take time. Check Photoshop memory. Try smaller batch.
Continuum tools missing	Action files must be in plugin folder. Reinstall if missing.
Colors look wrong	Verify correct palette selected. Check individual group blend modes.
Adjustment layers not added	Some layers may not get adjustments if histogram is unusual. Add manually.
Performance slow	Increase Photoshop memory allocation in Preferences. Close other applications.

12. Support & Contact

12.1 Getting Help

- **Website:** <https://pimagicstudio.com>
- **Tutorials:** <https://pimagicstudio.com/about/>
- **Email:** info@pimagicstudio.com
- **Include:** Processing log and screenshots when reporting issues

12.2 About the Creators

Dr. Christian Sasse and **Craig Stocks** created PiMagic to take the drudgery out of astrophotography post-processing so you can focus your time on the creative steps.

Craig Stocks operates **Utah Desert Remote Observatories** (utahdesertremote.com), providing remote telescope hosting under some of the darkest skies in the United States. The tutorials on pimagicstudio.com and the Utah Desert Remote Observatories YouTube channel draw from years of practical astrophotography experience.

Together they offer the **Southwest Utah Premium Astrophotography Masterclass** — a hands-on workshop combining iTelescope's premier equipment with expert processing instruction.

12.3 Purchase

Purchase PiMagic (includes both PiMagic and AstroMagic) at: <https://pimagicstudio.com/shop/>

— *Have Fun Processing!* —

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